

Activity using PostgreSQL Database

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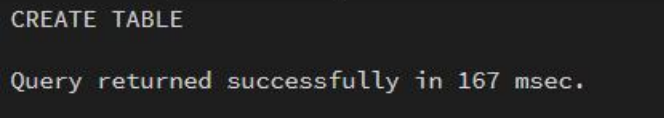
COURSE NAME:DSA0503-QUERY PROCESSING

1. Create the Department Table (Parent Table) with attributes
Department_ID INTEGER PRIMARY KEY, Department_Name TEXT

Code:

```
CREATE TABLE Department (  
    Department_ID INTEGER PRIMARY KEY,  
    Department_Name TEXT  
);
```

Ouput:



```
CREATE TABLE  
Query returned successfully in 167 msec.
```

2. Create the Student Table (Child Table) with the following attributes:
Student_ID INTEGER PRIMARY KEY,
Name TEXT,
Age INTEGER,
Department_ID INTEGER,
CGPA REAL,
FOREIGN KEY (Department_ID) REFERENCES
Department(Department_ID)

Code:

```
CREATE TABLE Student (  
    Student_ID INTEGER PRIMARY KEY,  
    Name TEXT,  
    Age INTEGER,  
    Department_ID INTEGER,  
    CGPA REAL,  
    FOREIGN KEY (Department_ID)  
    REFERENCES Department(Department_ID));
```

Output:

```
CREATE TABLE  
  
Query returned successfully in 167 msec.
```

3. Display students from CSE department.

Code:

```
SELECT s.*  
FROM Student s  
JOIN Department d  
ON s.Department_ID = d.Department_ID  
WHERE d.Department_Name = 'Computer Science';
```

Output:

	student_id [PK] integer	name text	age integer	department_id integer	cgpa real
1	101	Arun	20	1	8.5
2	102	Bala	21	1	9

4. Count students in each department.

Code:

```
SELECT Department_ID, COUNT(*) AS Student_Count  
FROM Student  
GROUP BY Department_ID;
```

Outupt:

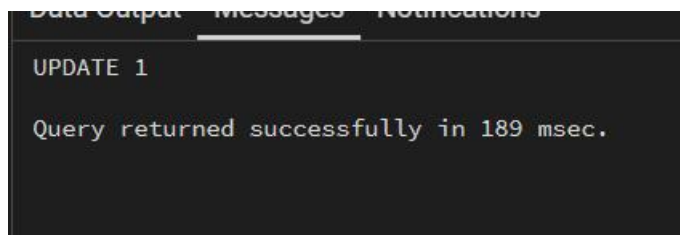
	department_id integer	student_count bigint
1	3	1
2	2	1
3	1	2

5. Update the foreign key. For eg., change the department of a student

Code:

```
UPDATE Student  
SET Department_ID = 2  
WHERE Student_ID = 101;
```

Output:

A screenshot of a SQL Developer window with tabs for 'Data Output', 'Messages', and 'Notifications'. The 'Messages' tab is active, showing the text 'UPDATE 1' and 'Query returned successfully in 189 msec.' in a monospaced font on a dark background.

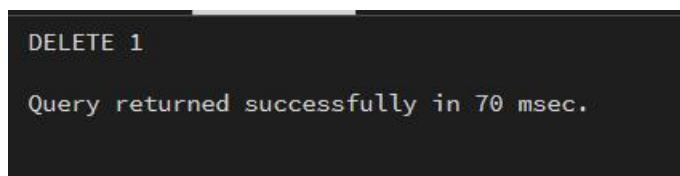
```
UPDATE 1  
Query returned successfully in 189 msec.
```

6. Delete a department

Code:

```
DELETE FROM Department  
WHERE Department_ID = 4;
```

Output:

A screenshot of a SQL Developer window with tabs for 'Data Output', 'Messages', and 'Notifications'. The 'Messages' tab is active, showing the text 'DELETE 1' and 'Query returned successfully in 70 msec.' in a monospaced font on a dark background.

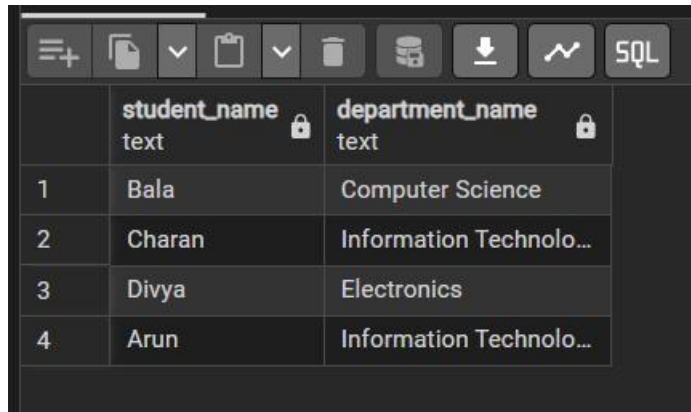
```
DELETE 1  
Query returned successfully in 70 msec.
```

7. Display each student's name along with the corresponding department name using an SQL JOIN.

Code:

```
SELECT  
    s.Name AS Student_Name,  
    d.Department_Name  
FROM Student s  
JOIN Department d  
ON s.Department_ID = d.Department_ID;
```

Output:



A screenshot of a database application interface. At the top is a toolbar with icons for menu, save, dropdown, clipboard, dropdown, delete, database, download, chart, and an 'SQL' button. Below the toolbar is a table with three columns: an index column, 'student_name' (text), and 'department_name' (text). The table contains four rows of data.

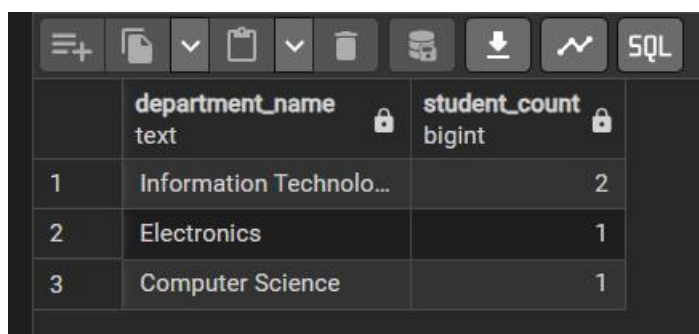
	student_name text	department_name text
1	Bala	Computer Science
2	Charan	Information Technolo...
3	Divya	Electronics
4	Arun	Information Technolo...

8. Display the number of students in each department.

Code:

```
SELECT
    d.Department_Name,
    COUNT(s.Student_ID) AS Student_Count
FROM Department d
LEFT JOIN Student s
ON d.Department_ID = s.Department_ID
GROUP BY d.Department_ID, d.Department_Name;
```

Output:



A screenshot of a database application interface, similar to the one above. The table has three columns: an index column, 'department_name' (text), and 'student_count' (bigint). The table contains three rows of data, showing the count of students for each department.

	department_name text	student_count bigint
1	Information Technolo...	2
2	Electronics	1
3	Computer Science	1

9. Display the names of students belonging to the Computer Science department.

Code:

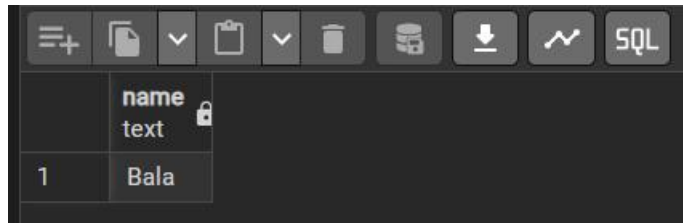
```
SELECT s.Name
FROM Student s
```

```

JOIN Department d
ON s.Department_ID = d.Department_ID
WHERE d.Department_Name = 'Computer Science';

```

Output:



The screenshot shows a database application interface with a toolbar at the top containing icons for menu, save, dropdown, clipboard, dropdown, delete, refresh, download, and a graph icon, followed by an 'SQL' button. Below the toolbar is a table with two columns: 'name' (text) and an empty column. The first row of the table contains the value '1' in the first column and 'Bala' in the second column.

	name text
1	Bala

10. Update the department of a student by changing the foreign key value and display the updated record.

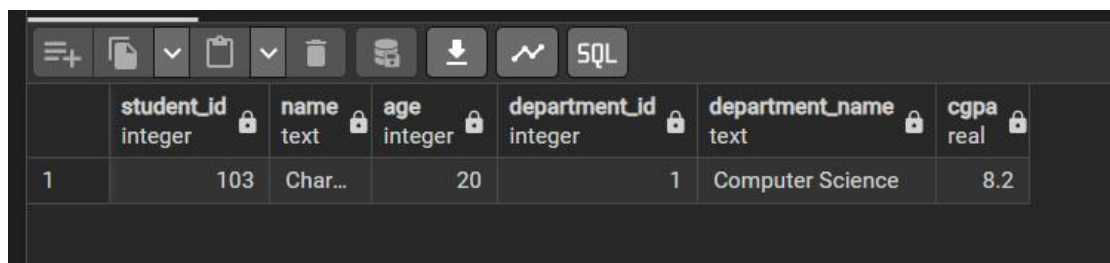
Code:

```

SELECT
    s.Student_ID,
    s.Name,
    s.Age,
    s.Department_ID,
    d.Department_Name,
    s.CGPA
FROM Student s
JOIN Department d
ON s.Department_ID = d.Department_ID
WHERE s.Student_ID = 103;

```

Output:



The screenshot shows a database application interface with a toolbar at the top containing icons for menu, save, dropdown, clipboard, dropdown, delete, refresh, download, and a graph icon, followed by an 'SQL' button. Below the toolbar is a table with seven columns: 'student_id' (integer), 'name' (text), 'age' (integer), 'department_id' (integer), 'department_name' (text), and 'cgpa' (real). The first row of the table contains the values 103, Char..., 20, 1, Computer Science, and 8.2.

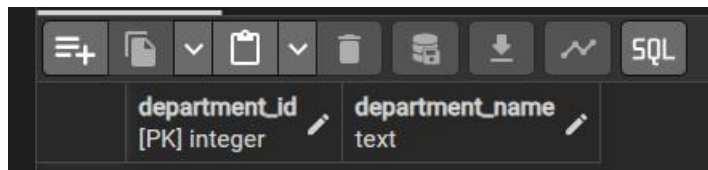
	student_id integer	name text	age integer	department_id integer	department_name text	cgpa real
1	103	Char...	20	1	Computer Science	8.2

11. Display departments that currently have no students assigned using an appropriate SQL JOIN.

Code:

```
SELECT
    d.Department_ID,
    d.Department_Name
FROM Department d
LEFT JOIN Student s
ON d.Department_ID = s.Department_ID
WHERE s.Student_ID IS NULL;
```

Output:



The screenshot shows a database management tool interface. At the top, there is a toolbar with various icons for file operations, database management, and SQL execution. Below the toolbar, a table structure is displayed with two columns: 'department_id' and 'department_name'. The 'department_id' column is marked as a primary key (PK) and has a data type of 'integer'. The 'department_name' column has a data type of 'text'. Both column names are underlined, indicating they are part of the table's schema.

<u>department_id</u>	<u>department_name</u>
[PK] integer	text